(Please write your Enrollment Number) Enrollment No:

MID-TERM EXAMINATION

(Course Name: B. Tech (ECE/ECE AI) (Semester: 3rd)

(October, 2025) OFF LINE mode

Subject Code: BEC 203	**Subject: Electronic Circuit Simulation Workshop**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**	**CO Mapping**
	a) Explain the h-parameter model of a transistor at low frequency.		CO1
	b) Compare CB and CE configurations of a transistor in terms of input resistance and current gain.		CO1
	c) State the significance of capacitances (Cπ and Cμ) in hybrid-π model.		CO2
	d) In a UJT relaxation oscillator, the value of R_{BB} is 5kΩ and stand off ratio η=0.6 at I_E=0. The value of R_{B2}.		CO2
	UNIT I		**CO Mapping**
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) Draw the hybrid-π equivalent circuit for a CE amplifier and explain each element.		CO1
	b) With the help of diagrams, explain the operation of a Darlington pair amplifier.		CO1
	c) A CE amplifier is drawn by a voltage source of internal resistance r_s = 800 Ω and the load resistance R_L=1000 Ω. The h-parameters are h_{ie}=1 kΩ, h_{re}=2x10^{-4}, h_{fe}=50 and h_{oe}=25μA/V. Compute the current gain (A_i), Input resistance (R_i), Voltage gain (A_v) and output resistance (R_o) using exact analysis and approximate analysis.		CO1
	UNIT II		**CO Mapping**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) Explain the Hybrid-π model at high frequency and its significance in amplifier		CO2
	b) Discuss the characteristics and applications of TRIAC and DIAC.		CO2
	c) Explain the working principle and operation of a Thyristor and list its practical applications.		CO2